

Introduction to linear regression

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The Human Freedom Index is a report that attempts to summarize the idea of “freedom” through a bunch of different variables for many countries around the globe. It serves as a rough objective measure for the relationships between the different types of freedom - whether it’s political, religious, economical or personal freedom - and other social and economic circumstances. The Human Freedom Index is an annually co-published report by the Cato Institute, the Fraser Institute, and the Liberales Institut at the Friedrich Naumann Foundation for Freedom.

In this lab, you’ll be analyzing data from Human Freedom Index reports from 2008-2016. Your aim will be to summarize a few of the relationships within the data both graphically and numerically in order to find which variables can help tell a story about freedom.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let’s load the packages.

```
library(tidyverse)
library(openintro)
data('hfi', package='openintro')
set.seed(1234)
```

The data

The data we’re working with is in the openintro package and it’s called **hfi**, short for Human Freedom Index.

1. What are the dimensions of the dataset?

Insert your answer here

```
#check dimesion
dim(hfi)
```

```
## [1] 1458 123
```

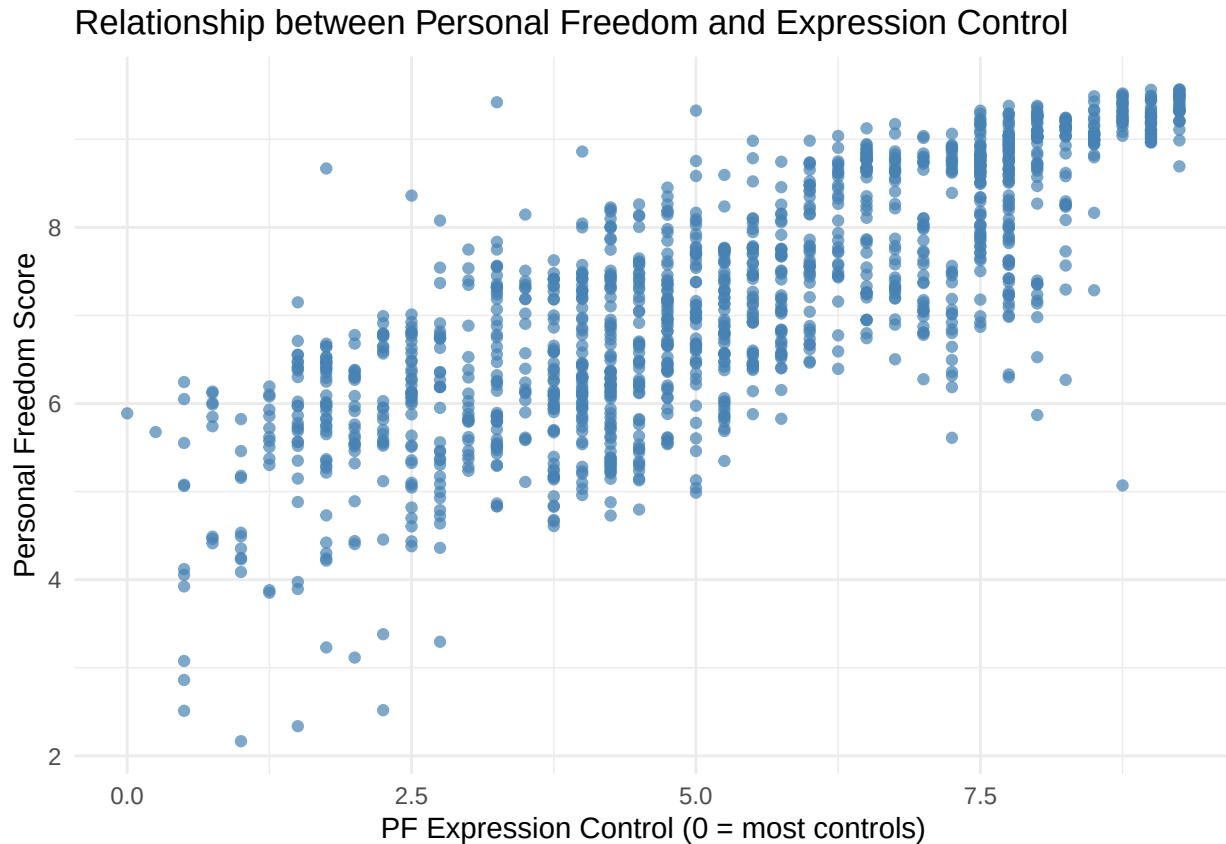
1458 rows and 123 columns

2. What type of plot would you use to display the relationship between the personal freedom score, **pf_score**, and one of the other numerical variables? Plot this relationship using the variable **pf_expression_control** as the predictor. Does the relationship look linear? If you knew a country’s **pf_expression_control**, or its score out of 10, with 0 being the most, of political pressures and controls on media content, would you be comfortable using a linear model to predict the personal freedom score?

Insert your answer here

Scatter plot to see the relationship between Personal Freedom and Expression Control

```
ggplot(hfi, aes(x = pf_expression_control, y = pf_score)) +
  geom_point(alpha = 0.7, color = "steelblue") +
  labs(x = "PF Expression Control (0 = most controls)",
       y = "Personal Freedom Score",
       title = "Relationship between Personal Freedom and Expression Control") +
  theme_minimal()
```



If the relationship looks linear, we can quantify the strength of the relationship with the correlation coefficient.

```
hfi %>%
  summarise(cor(pf_expression_control, pf_score, use = "complete.obs"))
```

```
## # A tibble: 1 x 1
##   `cor(pf_expression_control, pf_score, use = "complete.obs")`
##                                     <dbl>
## 1                                     0.796
```

The high correlation of 0.796 strongly supports using linear regression for prediction. Countries with greater media freedom (higher `pf_expression_control`) reliably have higher overall personal freedom scores.

Here, we set the `use` argument to “complete.obs” since there are some observations of NA.

Sum of squared residuals

In this section, you will use an interactive function to investigate what we mean by “sum of squared residuals”. You will need to run this function in your console, not in your markdown document. Running the function also requires that the `hfi` dataset is loaded in your environment.

Think back to the way that we described the distribution of a single variable. Recall that we discussed

characteristics such as center, spread, and shape. It's also useful to be able to describe the relationship of two numerical variables, such as `pf_expression_control` and `pf_score` above.

3. Looking at your plot from the previous exercise, describe the relationship between these two variables. Make sure to discuss the form, direction, and strength of the relationship as well as any unusual observations.

Insert your answer here

Relationship Description

The relationship appears to be roughly linear. The points generally follow a straight-line pattern rather than a curve or other nonlinear shape.

As PF expression control is 0 or around 0 there are very few data points of personal freedom score.

Direction: Positive relationship - As `pf_expression_control` increases (indicating fewer political pressures and controls on media content), the `pf_score` also increases. Countries with more media freedom tend to have higher overall personal freedom scores.

Strength: Strong relationship - With a correlation of 0.796, the points cluster fairly tightly around the imaginary line of best fit. This indicates that knowing a country's expression control score gives us good predictive power for estimating its personal freedom score.

Just as you've used the mean and standard deviation to summarize a single variable, you can summarize the relationship between these two variables by finding the line that best follows their association. Use the following interactive function to select the line that you think does the best job of going through the cloud of points.

```
# This will only work interactively (i.e. will not show in the knitted document)
hfi <- hfi %>% filter(complete.cases(pf_expression_control, pf_score))
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score)
```

After running this command, you'll be prompted to click two points on the plot to define a line. Once you've done that, the line you specified will be shown in black and the residuals in blue. Note that there are 30 residuals, one for each of the 30 observations. Recall that the residuals are the difference between the observed values and the values predicted by the line:

$$e_i = y_i - \hat{y}_i$$

The most common way to do linear regression is to select the line that minimizes the sum of squared residuals. To visualize the squared residuals, you can rerun the plot command and add the argument `showSquares = TRUE`.

```
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score, showSquares = TRUE)
```

Note that the output from the `plot_ss` function provides you with the slope and intercept of your line as well as the sum of squares.

4. Using `plot_ss`, choose a line that does a good job of minimizing the sum of squares. Run the function several times. What was the smallest sum of squares that you got? How does it compare to your neighbors?

Insert your answer here

Coefficients: (Intercept) x
4.8078 0.4267

Sum of Squares: 1011.625

The linear model

It is rather cumbersome to try to get the correct least squares line, i.e. the line that minimizes the sum of squared residuals, through trial and error. Instead, you can use the `lm` function in R to fit the linear model (a.k.a. regression line).

```
m1 <- lm(pf_score ~ pf_expression_control, data = hfi)
```

The first argument in the function `lm` is a formula that takes the form $y \sim x$. Here it can be read that we want to make a linear model of `pf_score` as a function of `pf_expression_control`. The second argument specifies that R should look in the `hfi` data frame to find the two variables.

The output of `lm` is an object that contains all of the information we need about the linear model that was just fit. We can access this information using the summary function.

```
summary(m1)
```

```
##
## Call:
## lm(formula = pf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8467 -0.5704  0.1452  0.6066  3.2060
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.61707    0.05745   80.36  <2e-16 ***
## pf_expression_control 0.49143    0.01006   48.85  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8318 on 1376 degrees of freedom
## (80 observations deleted due to missingness)
## Multiple R-squared:  0.6342, Adjusted R-squared:  0.634
## F-statistic: 2386 on 1 and 1376 DF, p-value: < 2.2e-16
```

Let's consider this output piece by piece. First, the formula used to describe the model is shown at the top. After the formula you find the five-number summary of the residuals. The "Coefficients" table shown next is key; its first column displays the linear model's y-intercept and the coefficient of `pf_expression_control`. With this table, we can write down the least squares regression line for the linear model:

$$\hat{y} = 4.61707 + 0.49143 \times pf_expression_control$$

One last piece of information we will discuss from the summary output is the Multiple R-squared, or more simply, R^2 . The R^2 value represents the proportion of variability in the response variable that is explained by the explanatory variable. For this model, 63.42% of the variability in `pf_free` is explained by `pf_expression_control`.

5. Fit a new model that uses `pf_expression_control` to predict `hf_score`, or the total human freedom score. Using the estimates from the R output, write the equation of the regression line. What does the slope tell us in the context of the relationship between human freedom and the amount of political pressure on media content?

Insert your answer here

```
m2 <- lm(hf_score ~ pf_expression_control, data = hfi)
summary(m2)
```

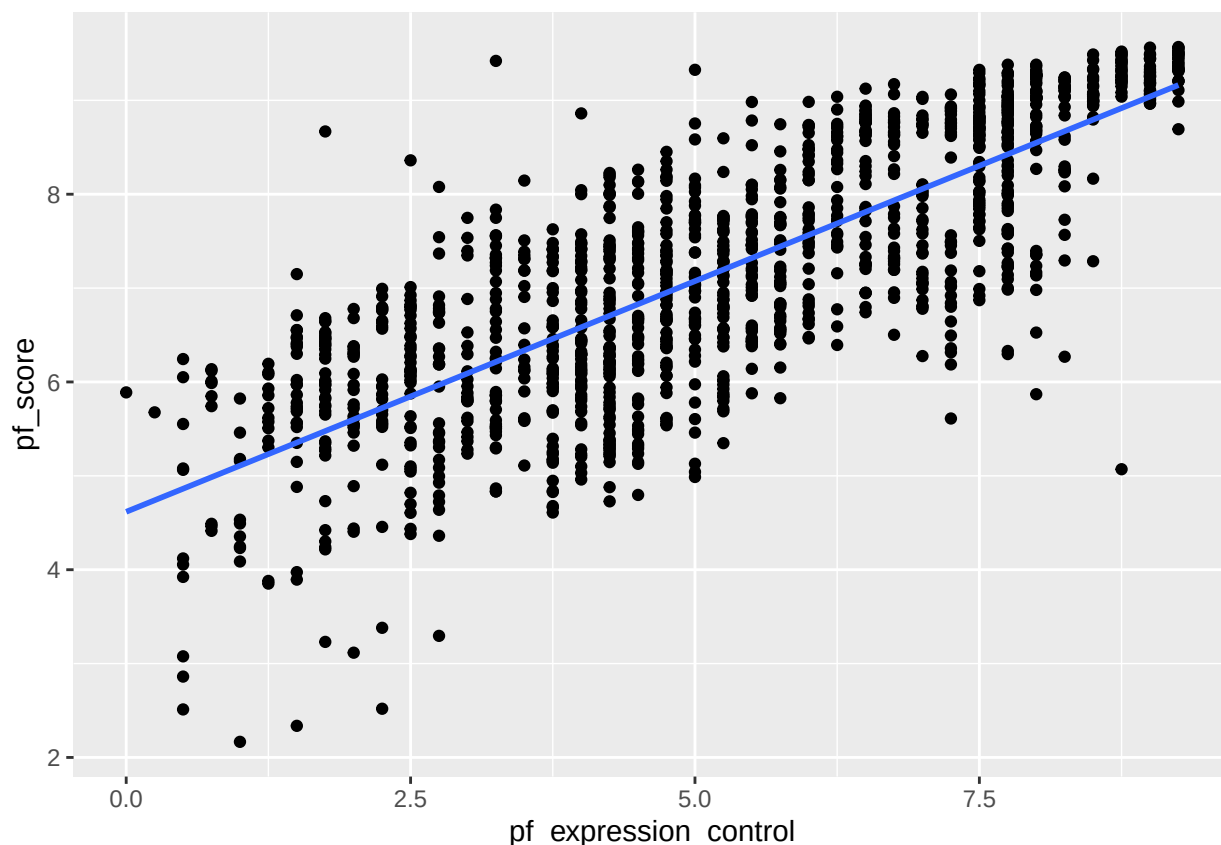
```
##
## Call:
## lm(formula = hf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.6198 -0.4908  0.1031  0.4703  2.2933
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.153687   0.046070  111.87  <2e-16 ***
## pf_expression_control 0.349862   0.008067   43.37  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.667 on 1376 degrees of freedom
## (80 observations deleted due to missingness)
## Multiple R-squared:  0.5775, Adjusted R-squared:  0.5772
## F-statistic: 1881 on 1 and 1376 DF, p-value: < 2.2e-16
```

$$\hat{y} = 5.153687 + 0.349862 \times pf_expression_control$$

Prediction and prediction errors

Let's create a scatter plot with the least squares line for m1 laid on top.

```
ggplot(data = hfi, aes(x = pf_expression_control, y = pf_score)) +
  geom_point() +
  stat_smooth(method = "lm", se = FALSE)
```



Here, we are literally adding a layer on top of our plot. `geom_smooth` creates the line by fitting a linear model. It can also show us the standard error `se` associated with our line, but we'll suppress that for now.

This line can be used to predict y at any value of x . When predictions are made for values of x that are beyond the range of the observed data, it is referred to as *extrapolation* and is not usually recommended. However, predictions made within the range of the data are more reliable. They're also used to compute the residuals.

6. If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom school for one with a 6.7 rating for `pf_expression_control`? Is this an overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

Insert your answer here

for 6.7 `pf_expression_control` it seems like 7.9 `pf_score` on the line when we put it in the formula we get:

$$\hat{y} = 4.61707 + 0.49143 \times pf_expression_control$$

7.91

```
# Find the country with pf_expression_control closest to 6.7
```

```
country_6_7 <- hfi %>%
  filter(!is.na(pf_expression_control), !is.na(pf_score)) %>%
  mutate(diff = abs(pf_expression_control - 6.7)) %>%
  arrange(diff) %>%
  head(1)
```

```
# Calculate predicted value and residual
predicted <- 4.61707 + 0.49143 * country_6_7$pf_expression_control
residual <- country_6_7$pf_score - predicted
```

```
country_6_7$pf_expression_control
```

```
## [1] 6.75
```

```
country_6_7$pf_score
```

```
## [1] 7.430864
```

```
residual
```

```
## [1] -0.5033587
```

If someone saw only the least squares regression line, they would predict a personal freedom score of approximately 7.9 for a country with `pf_expression_control = 6.7`

This prediction is an OVERESTIMATE by approximately 0.503 points.

The negative residual of -0.503 indicates that:

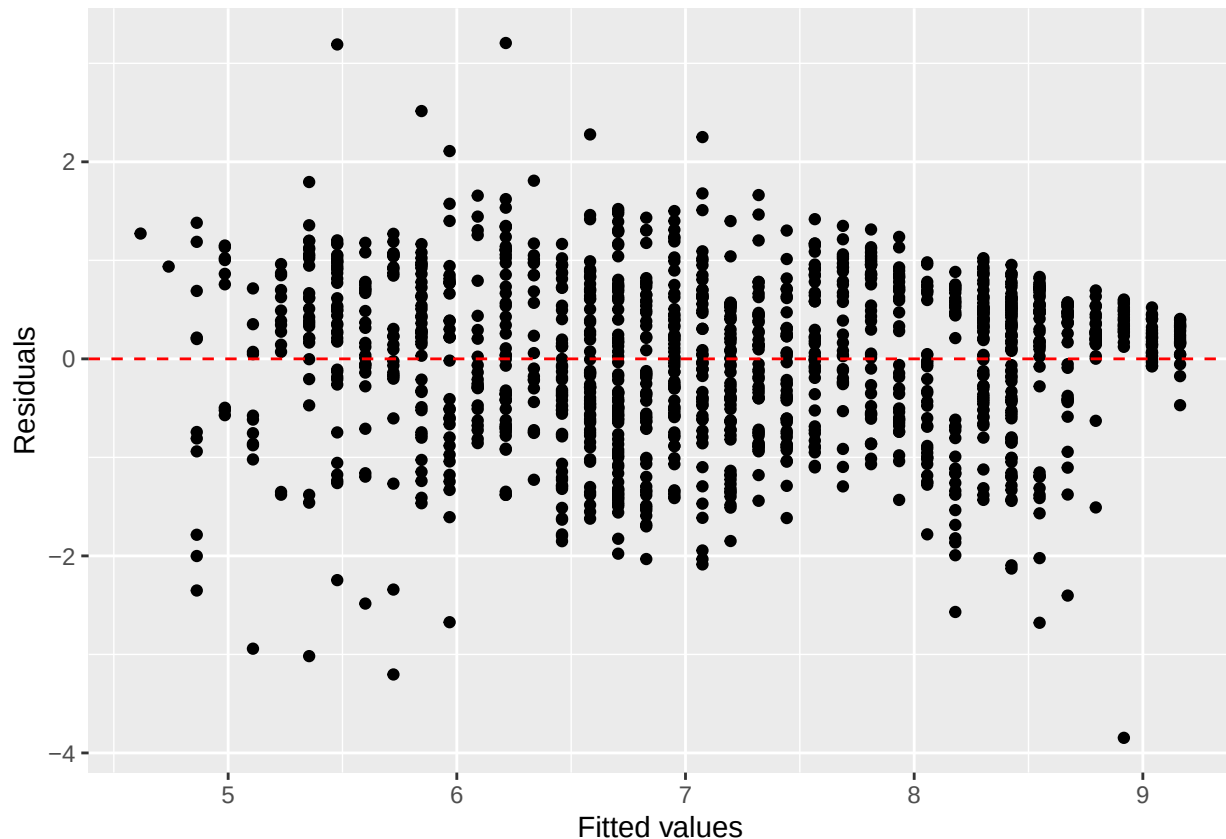
The actual personal freedom score (7.43) is lower than what the regression line predicts (7.9)

Model diagnostics

To assess whether the linear model is reliable, we need to check for (1) linearity, (2) nearly normal residuals, and (3) constant variability.

Linearity: You already checked if the relationship between `pf_score` and ‘`pf_expression_control`’ is linear using a scatterplot. We should also verify this condition with a plot of the residuals vs. fitted (predicted) values.

```
ggplot(data = m1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") + # changed line color to red
  xlab("Fitted values") +
  ylab("Residuals")
```



Notice here that `m1` can also serve as a data set because stored within it are the fitted values ($\hat{y}$) and the residuals. Also note that we're getting fancy with the code here. After creating the scatterplot on the first layer (first line of code), we overlay a horizontal dashed line at $y = 0$ (to help us check whether residuals are distributed around 0), and we also rename the axis labels to be more informative.

7. Is there any apparent pattern in the residuals plot? What does this indicate about the linearity of the relationship between the two variables?

Insert your answer here

- Pattern in Residuals Plot:

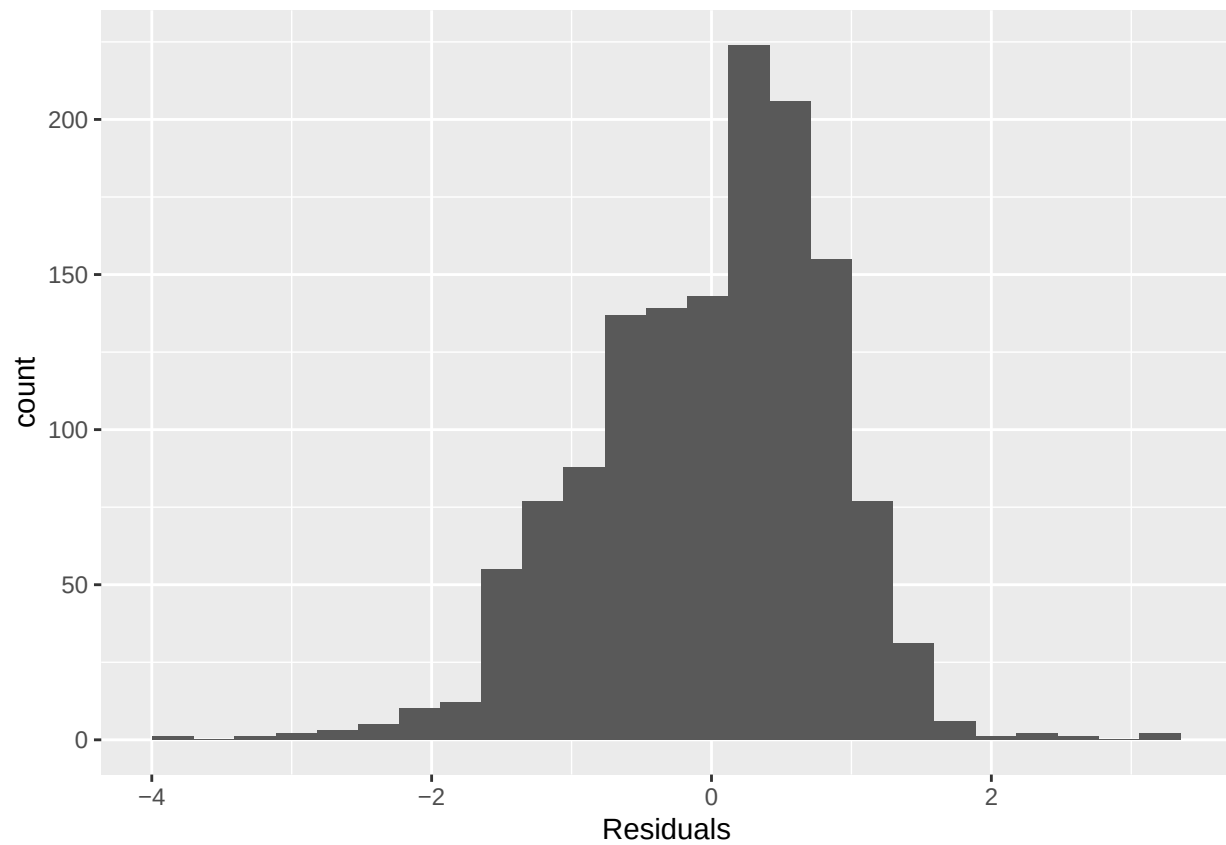
The residuals appear to be randomly scattered around the horizontal line at $y=0$ with no strong systematic pattern. The points are fairly evenly distributed above and below the zero line across the range of fitted values.

Evidence Supporting Linearity:

- Random scatter - No obvious curved pattern or systematic structure
- Even distribution - Residuals are balanced above and below zero across all fitted values
- No clear trends - The spread doesn't systematically increase or decrease

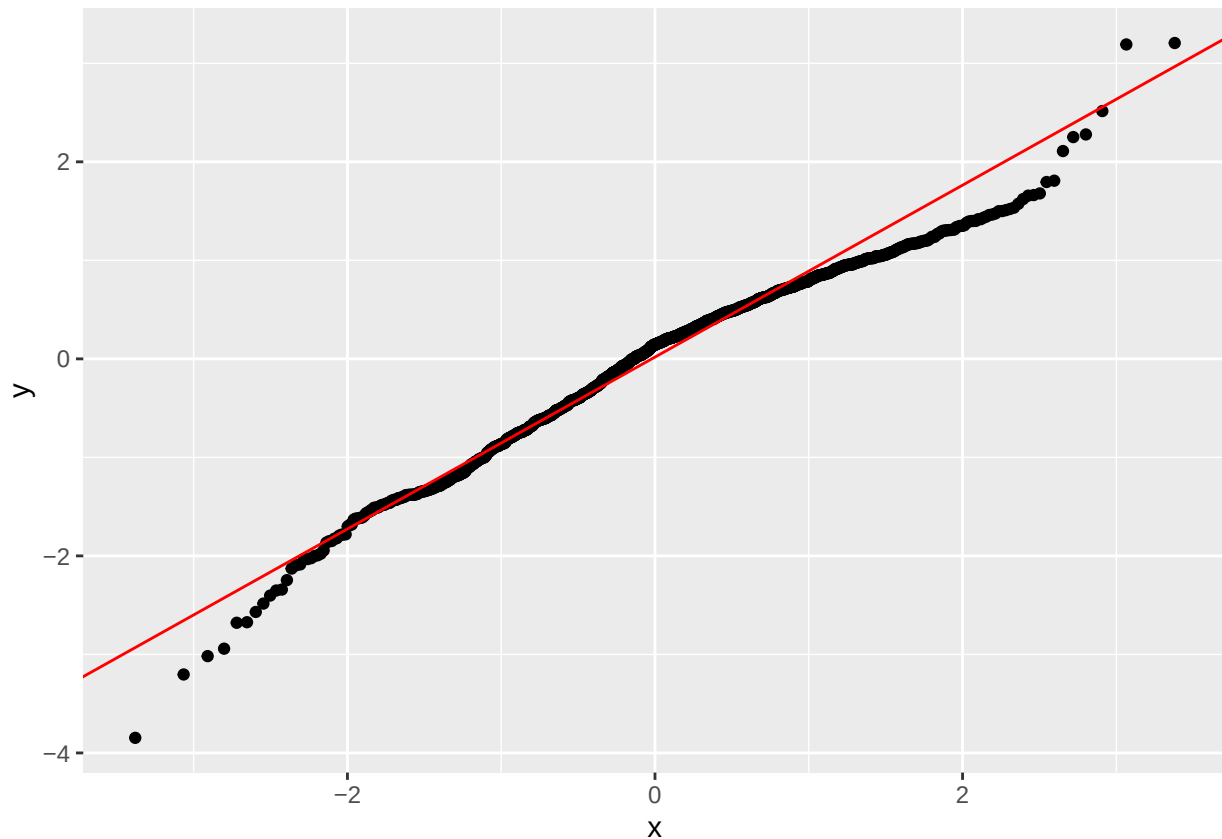
Nearly normal residuals: To check this condition, we can look at a histogram

```
ggplot(data = m1, aes(x = .resid)) +
  geom_histogram(bins = 25) +
  xlab("Residuals")
```

or a normal probability plot of the residuals.

```
ggplot(data = m1, aes(sample = .resid)) +  
  stat_qq() +  
  stat_qq_line(color='red') # reference line has been added
```



Note that the syntax for making a normal probability plot is a bit different than what you're used to seeing: we set `sample` equal to the residuals instead of `x`, and we set a statistical method `qq`, which stands for “quantile-quantile”, another name commonly used for normal probability plots.

8. Based on the histogram and the normal probability plot, does the nearly normal residuals condition appear to be met?

Insert your answer here Based on the histogram and normal probability plot, the nearly normal residuals condition appears to be met.

The histogram shows that the majority of residual values are concentrated within the range of -2 to 2, with a clear center around 0, indicating a roughly symmetric, bell-shaped distribution.

The normal probability (Q-Q) plot further confirms this condition, as most data points fall closely along the reference line, demonstrating that the residuals follow an approximately normal distribution. This linear pattern in the Q-Q plot satisfies the nearly normal residuals assumption required for reliable linear regression inference.

Constant variability:

9. Based on the residuals vs. fitted plot, does the constant variability condition appear to be met?

Insert your answer here

Based on the residuals vs. fitted plot, the constant variability condition appears to be reasonably met.

The spread of the residuals shows relatively consistent variability across all ranges of the fitted values. There is no clear fanning pattern where the spread systematically increases or decreases as the fitted values change. The residuals maintain approximately the same vertical spread throughout the horizontal axis, which satisfies the constant variability assumption for reliable linear regression.

More Practice

- Choose another freedom variable and a variable you think would strongly correlate with it.. Produce a scatterplot of the two variables and fit a linear model. At a glance, does there seem to be a linear relationship?

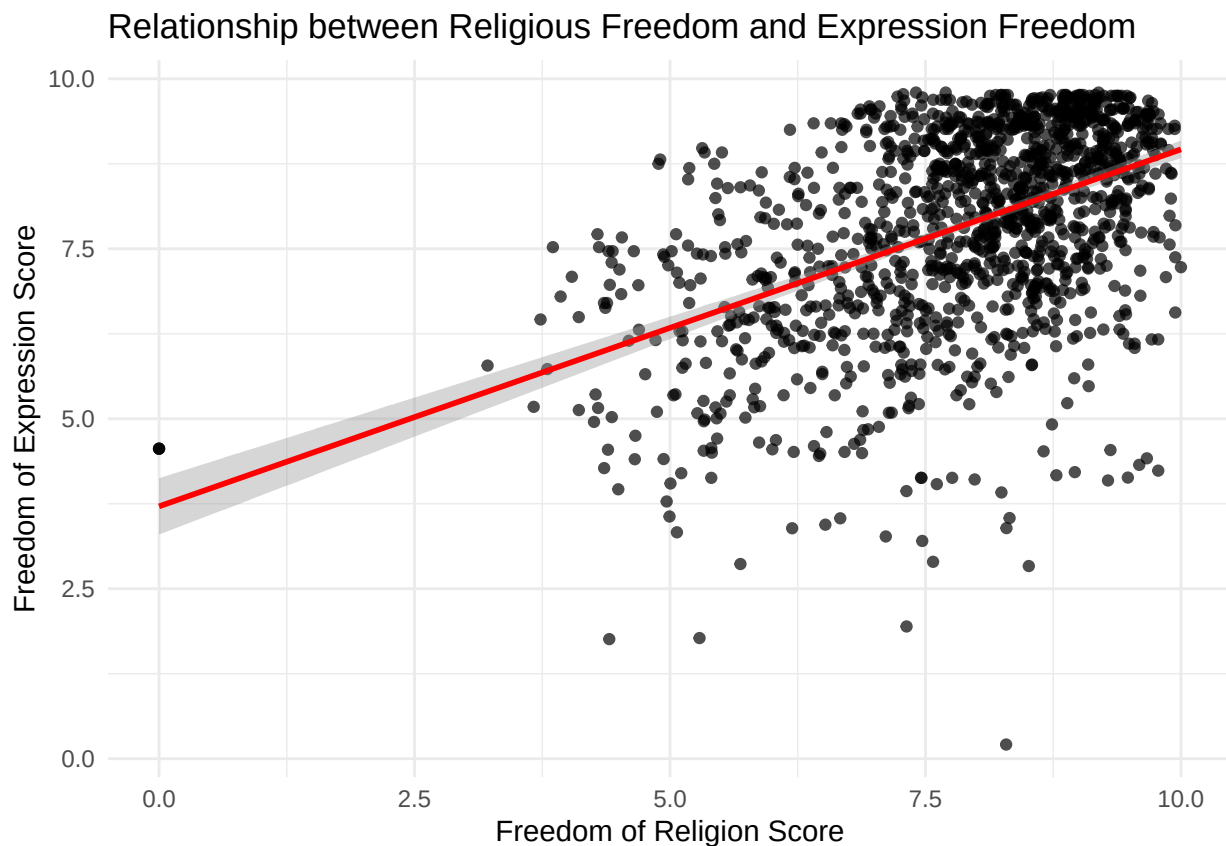
Insert your answer here

```
colnames(hfi) # check columns

##      [1] "year"                "ISO_code"
##      [3] "countries"           "region"
##      [5] "pf_rol_procedural"    "pf_rol_civil"
##      [7] "pf_rol_criminal"     "pf_rol"
##      [9] "pf_ss_homicide"       "pf_ss_disappearances_disap"
##     [11] "pf_ss_disappearances_violent" "pf_ss_disappearances_organized"
##     [13] "pf_ss_disappearances_fatalities" "pf_ss_disappearances_injuries"
##     [15] "pf_ss_disappearances"    "pf_ss_women_fgm"
##     [17] "pf_ss_women_missing"     "pf_ss_women_inheritance_widows"
##     [19] "pf_ss_women_inheritance_daughters" "pf_ss_women_inheritance"
##     [21] "pf_ss_women"             "pf_ss"
##     [23] "pf_movement_domestic"    "pf_movement_foreign"
##     [25] "pf_movement_women"       "pf_movement"
##     [27] "pf_religion_estop_establish" "pf_religion_estop_operate"
##     [29] "pf_religion_estop"       "pf_religion_harassment"
##     [31] "pf_religion_restrictions" "pf_religion"
##     [33] "pf_association_association" "pf_association_assembly"
##     [35] "pf_association_political_establish" "pf_association_political_operate"
##     [37] "pf_association_political" "pf_association_prof_establish"
##     [39] "pf_association_prof_operate" "pf_association_prof"
##     [41] "pf_association_sport_establish" "pf_association_sport_operate"
##     [43] "pf_association_sport"    "pf_association"
##     [45] "pf_expression_killed"    "pf_expression_jailed"
##     [47] "pf_expression_influence" "pf_expression_control"
##     [49] "pf_expression_cable"     "pf_expression_newspapers"
##     [51] "pf_expression_internet"  "pf_expression"
##     [53] "pf_identity_legal"       "pf_identity_parental_marriage"
##     [55] "pf_identity_parental_divorce" "pf_identity_parental"
##     [57] "pf_identity_sex_male"    "pf_identity_sex_female"
##     [59] "pf_identity_sex"        "pf_identity_divorce"
##     [61] "pf_identity"            "pf_score"
##     [63] "pf_rank"                "ef_government_consumption"
##     [65] "ef_government_transfers" "ef_government_enterprises"
##     [67] "ef_government_tax_income" "ef_government_tax_payroll"
##     [69] "ef_government_tax"      "ef_government"
##     [71] "ef_legal_judicial"      "ef_legal_courts"
##     [73] "ef_legal_protection"    "ef_legal_military"
##     [75] "ef_legal_integrity"     "ef_legal_enforcement"
##     [77] "ef_legal_restrictions"  "ef_legal_police"
##     [79] "ef_legal_crime"         "ef_legal_gender"
##     [81] "ef_legal"               "ef_money_growth"
##     [83] "ef_money_sd"            "ef_money_inflation"
##     [85] "ef_money_currency"      "ef_money"
##     [87] "ef_trade_tariffs_revenue" "ef_trade_tariffs_mean"
##     [89] "ef_trade_tariffs_sd"    "ef_trade_tariffs"
##     [91] "ef_trade_regulatory_nontariff" "ef_trade_regulatory_compliance"
```

```
## [93] "ef_trade_regulatory"      "ef_trade_black"
## [95] "ef_trade_movement_foreign" "ef_trade_movement_capital"
## [97] "ef_trade_movement_visit"  "ef_trade_movement"
## [99] "ef_trade"                 "ef_regulation_credit_ownership"
## [101] "ef_regulation_credit_private" "ef_regulation_credit_interest"
## [103] "ef_regulation_credit"      "ef_regulation_labor_minwage"
## [105] "ef_regulation_labor_firing" "ef_regulation_labor_bargain"
## [107] "ef_regulation_labor_hours"  "ef_regulation_labor_dismissal"
## [109] "ef_regulation_labor_conscription" "ef_regulation_labor"
## [111] "ef_regulation_business_adm"  "ef_regulation_business_bureaucracy"
## [113] "ef_regulation_business_start" "ef_regulation_business_bribes"
## [115] "ef_regulation_business_licensing" "ef_regulation_business_compliance"
## [117] "ef_regulation_business"      "ef_regulation"
## [119] "ef_score"                    "ef_rank"
## [121] "hf_score"                    "hf_rank"
## [123] "hf_quartile"
```

```
# Create scatterplot with linear model
ggplot(hfi, aes(x = pf_religion, y = pf_expression)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(x = "Freedom of Religion Score",
       y = "Freedom of Expression Score",
       title = "Relationship between Religious Freedom and Expression Freedom") +
  theme_minimal()
```



```
# Fit linear model
m_religion_expression <- lm(pf_expression ~ pf_religion, data = hfi)
```

```
summary(m_religion_expression)
```

```
##
## Call:
## lm(formula = pf_expression ~ pf_religion, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.8555 -0.7610  0.2158  1.0185  2.5262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.71116    0.21113   17.58  <2e-16 ***
## pf_religion  0.52509    0.02643   19.87  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.318 on 1366 degrees of freedom
## (90 observations deleted due to missingness)
## Multiple R-squared:  0.2242, Adjusted R-squared:  0.2237
## F-statistic: 394.8 on 1 and 1366 DF,  p-value: < 2.2e-16
```

- How does this relationship compare to the relationship between `pf_expression_control` and `pf_score`? Use the R^2 values from the two model summaries to compare. Does your independent variable seem to predict your dependent one better? Why or why not?

Insert your answer here

```
# Model 1: pf_expression_control predicting pf_score
m1 <- lm(pf_score ~ pf_expression_control, data = hfi)
summary(m1)$r.squared
```

```
## [1] 0.6342361
```

```
# Model 2: pf_religion predicting pf_expression
m_religion_expression <- lm(pf_expression ~ pf_religion, data = hfi)
summary(m_religion_expression)
```

```
##
## Call:
## lm(formula = pf_expression ~ pf_religion, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.8555 -0.7610  0.2158  1.0185  2.5262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.71116    0.21113   17.58  <2e-16 ***
## pf_religion  0.52509    0.02643   19.87  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.318 on 1366 degrees of freedom
## (90 observations deleted due to missingness)
## Multiple R-squared:  0.2242, Adjusted R-squared:  0.2237
```

F-statistic: 394.8 on 1 and 1366 DF, p-value: < 2.2e-16

R² Comparison:

Model 1 (pf_expression_control → pf_score): $R^2 = 0.634$ (63.4%) Model 2 (pf_religion → pf_expression): $R^2 = 0.224$ (22.4%) Conclusion:

pf_expression_control predicts pf_score MUCH better than pf_religion predicts pf_expression.

Why the Big Difference:

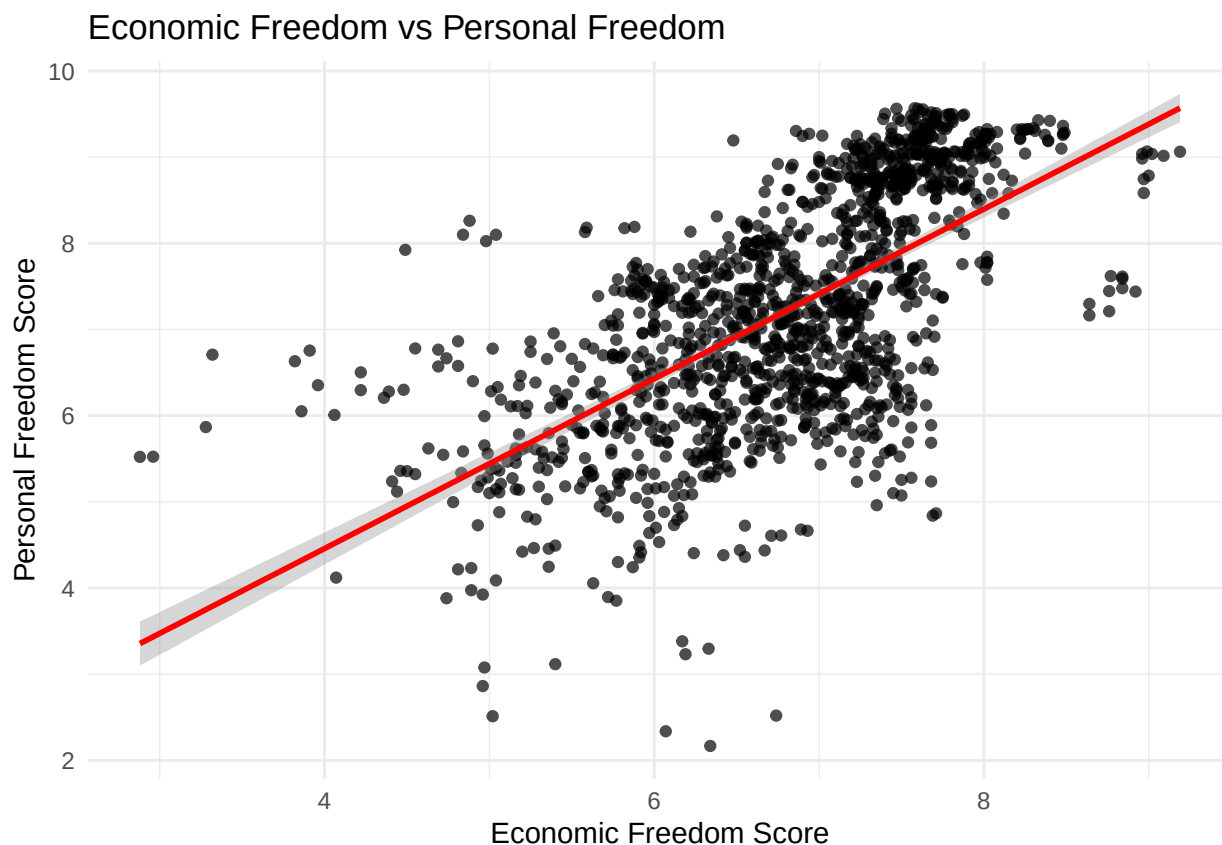
Stronger Direct Relationship: Expression control is a direct component of the personal freedom score, while religious freedom and expression freedom are more distantly related concepts Better Explanatory Power:

- Model 1 explains 63.4% of the variation in personal freedom scores
- Model 2 explains only 22.4% of the variation in expression freedom scores Conceptual Closeness: A country's controls on media content (pf_expression_control) are more directly tied to its overall personal freedom rating than its religious freedom policies are to its specific expression freedom score
- What's one freedom relationship you were most surprised about and why? Display the model diagnostics for the regression model analyzing this relationship.

Insert your answer here Economic Freedom vs. Personal Freedom

I'd examine the relationship between ef_score (economic freedom) and pf_score (personal freedom).

```
# Scatterplot of economic vs personal freedom
ggplot(hfi, aes(x = ef_score, y = pf_score)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", color = "red") +
  labs(x = "Economic Freedom Score",
       y = "Personal Freedom Score",
       title = "Economic Freedom vs Personal Freedom") +
  theme_minimal()
```

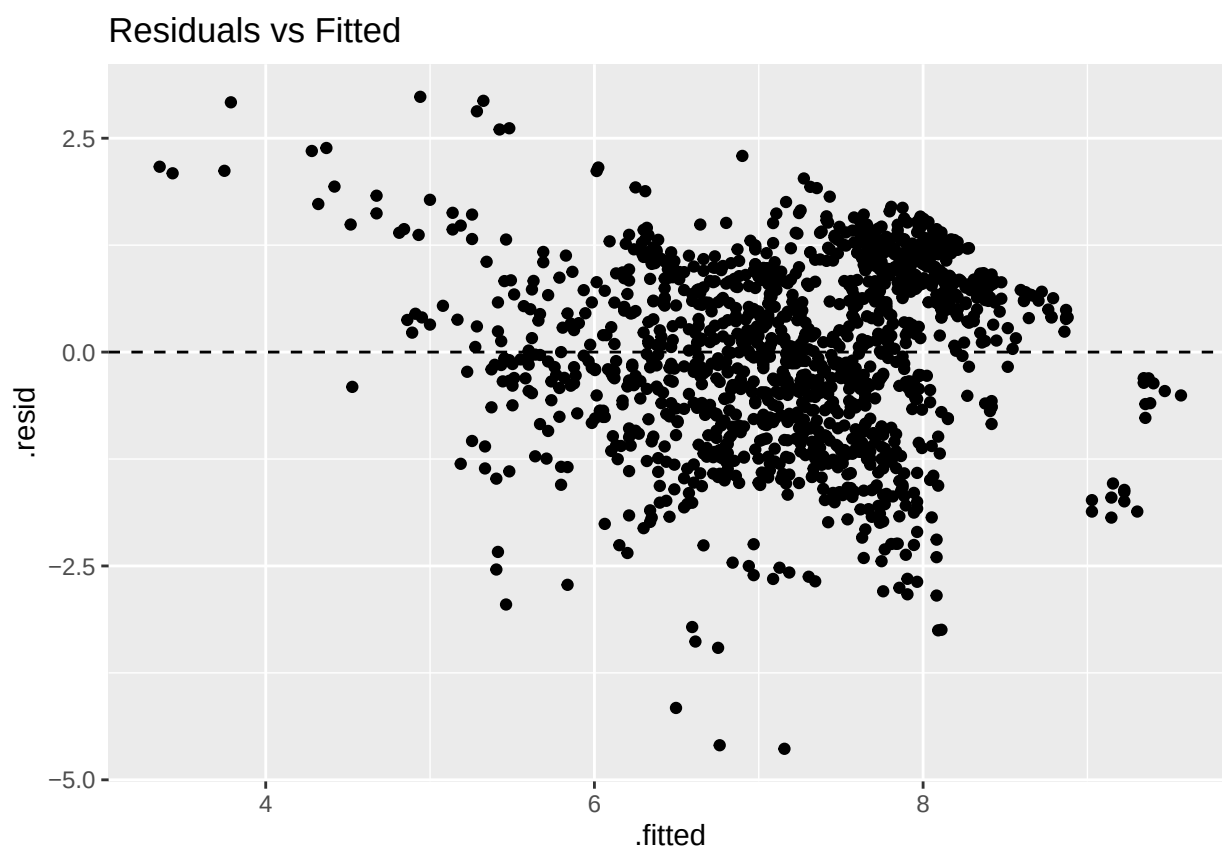


```
# Fit linear model
m_econ_personal <- lm(pf_score ~ ef_score, data = hfi)
summary(m_econ_personal)

##
## Call:
## lm(formula = pf_score ~ ef_score, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.6373 -0.7334  0.1090  0.8353  2.9840
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.51812    0.22223   2.331  0.0199 *
## ef_score     0.98490    0.03248  30.326 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.065 on 1376 degrees of freedom
## (80 observations deleted due to missingness)
## Multiple R-squared:  0.4006, Adjusted R-squared:  0.4002
## F-statistic: 919.7 on 1 and 1376 DF, p-value: < 2.2e-16

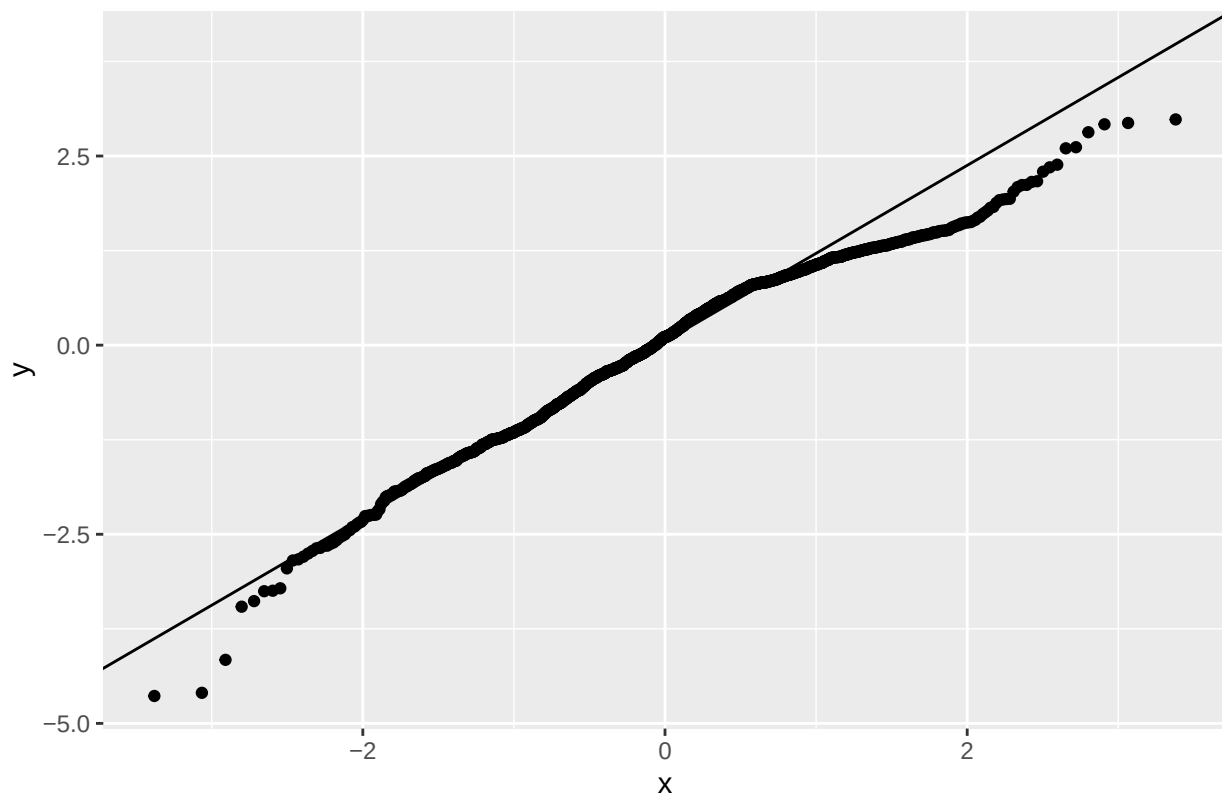
# Residuals vs Fitted
ggplot(m_econ_personal, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
```

```
labs(title = "Residuals vs Fitted")
```

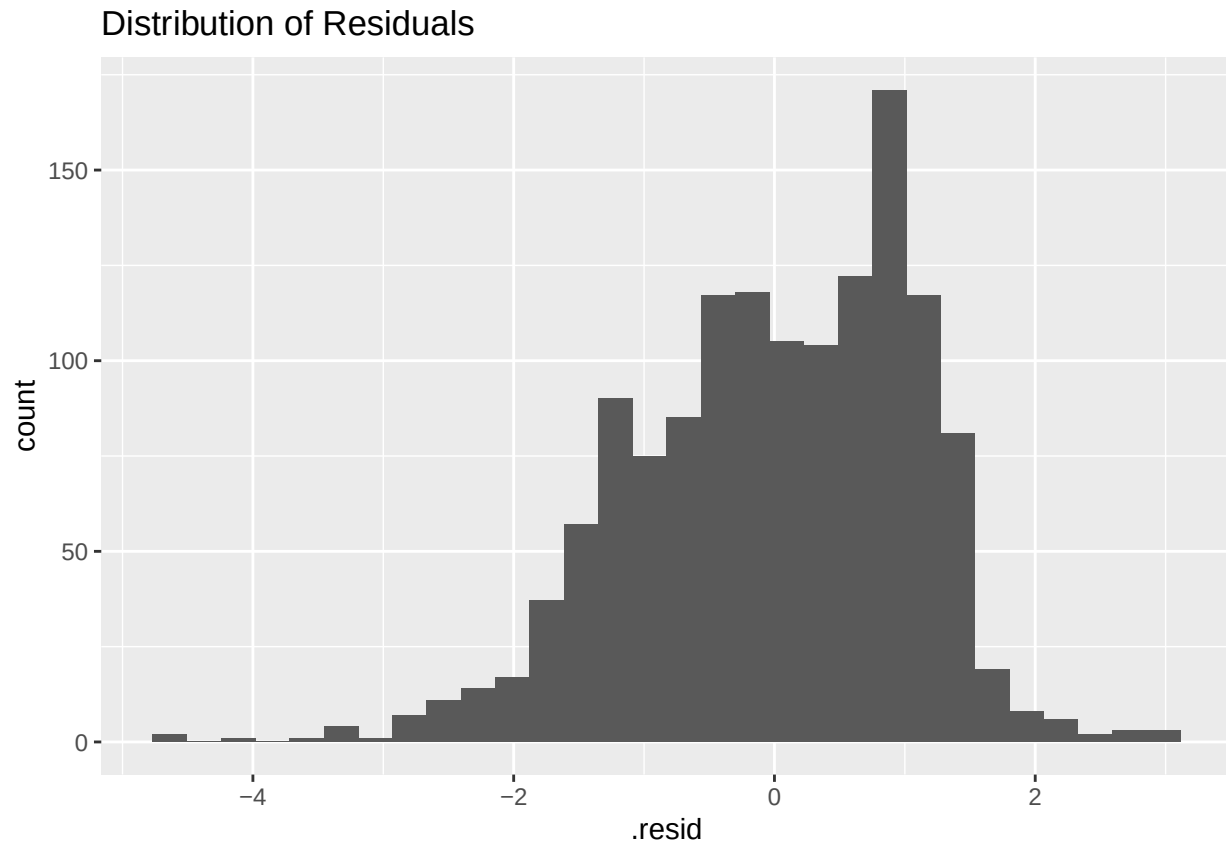


```
# Q-Q Plot for normality  
ggplot(m_econ_personal, aes(sample = .resid)) +  
  stat_qq() +  
  stat_qq_line() +  
  labs(title = "Q-Q Plot of Residuals")
```


Q-Q Plot of Residuals



```
# Histogram of residuals  
ggplot(m_econ_personal, aes(x = .resid)) +  
  geom_histogram(bins = 30) +  
  labs(title = "Distribution of Residuals")
```



Key Findings: Strength of Relationship:

$R^2 = 0.401$ - Economic freedom explains 40.1% of the variation in personal freedom scores. This is moderately strong but weaker than the expression control relationship ($R^2 = 0.634$).

Why This Might Be Surprising:

Not as Strong as Expected: While positive, the relationship ($R^2 = 0.401$) is weaker than some might assume - economic freedom doesn't perfectly predict personal freedom.

Residual Range: Residuals from -4.64 to +2.98 show substantial variation unexplained by economic freedom alone. This reveals that while economic and personal freedoms are related, they're not inseparable - some countries prioritize one over the other, creating fascinating case studies in governance approaches.
